13 Lecture 13 (February 12, 2025, est.): Completing Tannaka reconstruction

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We now restore the symmetric tensor structure on $\mathcal{C}$. Let $\omega : \mathcal{C} \to \mathrm{Vec}_k$ be a symmetric tensor functor. The preceding lectures construct a coalgebra B and an equivalence

$$\mathcal{C} \simeq \mathrm{Comod}_{\mathrm{fd}}(B)$$

compatible with the underlying vector spaces. We show that the tensor product defines multiplication on B, symmetry makes it commutative, and rigidity supplies the antipode.

Tensor products of comodules

Suppose first that B is merely a coalgebra. A coalgebra homomorphism $m : B \otimes B \to B$ equips the vector space $V \otimes W$ with the coaction

$$v \otimes w \longmapsto \sum v_{(0)} \otimes w_{(0)} \otimes m(v_{(1)} \otimes w_{(1)}).$$

Coassociativity and the counit identity follow from the fact that m is a coalgebra map. This construction is functorial and bilinear on morphisms. Conversely, a bilinear bifunctor on finite-dimensional B-comodules whose underlying vector-space functor is the ordinary tensor product determines a unique such m. Explicitly, the tensor product of two matrix coefficients is sent to the corresponding coefficient of the specified coaction on $V \otimes W$. The naturality relations of Lecture 12, separately in the two variables, make this well-defined, and the coefficient coproduct formula makes it a coalgebra homomorphism.

The usual vector-space associator is a comodule map exactly when m is associative: on a triple tensor product the two coactions have coefficients $m(m(b \otimes c) \otimes d)$ and $m(b \otimes m(c \otimes d))$. Since coefficients span B, equality on all triple tensor products is equality of those maps. Similarly, the ordinary flip is a comodule map exactly when m is commutative. A unit object with underlying space k determines a coalgebra map $u : k \to B$; the usual unit constraints are comodule maps precisely when u is a two-sided unit for m.

A lemma about rigidity

13.1 Lemma. *Let $\mathcal{C}$ be rigid and let $F, G : \mathcal{C} \to \mathcal{D}$ be strong monoidal functors. Every monoidal natural transformation $\alpha : F \to G$ is invertible.*

Proof. Strong monoidal functors preserve duals. Use the induced identifications $F(X^*) \simeq F(X)^*$ and $G(X^*) \simeq G(X)^*$, and suppress associators and unit constraints. Define

$$\begin{aligned}\beta_X : G(X) &\xrightarrow{\mathrm{coev}_{F(X)} \otimes \mathrm{id}} F(X) \otimes F(X)^* \otimes G(X) \\ &\xrightarrow{\mathrm{id} \otimes \alpha_{X^*} \otimes \mathrm{id}} F(X) \otimes G(X)^* \otimes G(X) \xrightarrow{\mathrm{id} \otimes \mathrm{ev}_{G(X)}} F(X).\end{aligned}$$

Naturality with evaluation, together with the tensor and unit identities for α, gives

$$\mathrm{ev}_{G(X)}(\alpha_{X^*} \otimes \alpha_X) = \mathrm{ev}_{F(X)}\,.$$

It follows from the first triangle identity that $\beta_X \alpha_X = \mathrm{id}_{F(X)}$. Naturality with coevaluation gives

$$(\alpha_X \otimes \alpha_{X^*})\,\mathrm{coev}_{F(X)} = \mathrm{coev}_{G(X)},$$

so the other composite is $\alpha_X \beta_X = \mathrm{id}_{G(X)}$. Thus all components are invertible, and their inverses automatically form a natural monoidal transformation. ∎

No rigidity assumption on every object of $\mathcal{D}$ is needed: the objects in the images of the two functors already have the required duals.

The reconstructed Hopf algebra

We prove Theorem 10.4. Write matrix coefficients in the notation of Lecture 12, and use the monoidal identifications of ω to suppress its tensor constraints. Define

$$[\phi \otimes v]_X[\psi \otimes w]_Y = [(\phi \otimes \psi) \otimes (v \otimes w)]_{X\otimes Y}, \qquad 1_B = [1^* \otimes 1]_{\mathbb{1}},$$

where $1^* : k \to k$ is the identity. Naturality of the tensor product respects the coefficient relations, so multiplication is well-defined. The associator and unit constraints give associativity and the unit identities. The symmetric braiding maps under ω to the ordinary flip, so its naturality relation identifies the two orders of the product. Thus B is a commutative algebra.

Using tensor products of bases in the coefficient formula gives

$$\Delta(bc) = \Delta(b)\Delta(c), \qquad \varepsilon(bc) = \varepsilon(b)\varepsilon(c),$$

and $\Delta(1_B) = 1_B \otimes 1_B$, $\varepsilon(1_B) = 1$. Therefore B is a commutative bialgebra. By construction, its comodule tensor product agrees with the one transported from $\mathcal{C}$.

For each commutative k-algebra R, Lecture 12 identifies a linear map $g : B \to R$ with the natural endomorphism

$$\lambda_X(v \otimes 1) = \sum_i e_i \otimes g([e_i^* \otimes v]_X)$$

of ω_R. The multiplication and unit formulas show that

$$\lambda_{X\otimes Y} = \lambda_X \otimes_R \lambda_Y, \quad \lambda_{\mathbb{1}} = \mathrm{id}_R \quad \Longleftrightarrow \quad g(bc) = g(b)g(c), \quad g(1_B) = 1_R.$$

Consequently,

$$\underline{\mathrm{End}}^{\otimes}(\omega)(R) \simeq \mathrm{Hom}_{k\text{-alg}}(B, R).$$

Rigidity and Lemma 13.1, applied to ω_R, show that every such tensor endomorphism is invertible. The functor represented by B is therefore a group functor, and its inverse is represented by an antipode $S : B \to B$.

One can write the antipode explicitly. Identify $\omega(X^*)$ with $(\omega X)^*$, and for $v \in \omega X$ let $\widehat{v} \in (\omega X)^{**}$ be evaluation at v. Then

$$S([\phi \otimes v]_X) = [\widehat{v} \otimes \phi]_{X^*}.$$

The evaluation and coevaluation identities give

$$m(S \otimes \mathrm{id})\Delta = u\varepsilon = m(\mathrm{id} \otimes S)\Delta,$$

which are also the identities expressing that the action on the dual is the inverse transpose. This realizes the group-functor inverse just constructed.

Set $G = \operatorname{Spec} B$. We have proved, naturally in R,

$$G(R) = \underline{\mathrm{Aut}}^{\otimes}(\omega)(R).$$

Equation (12.1) now becomes

$$\widetilde{\omega} : \mathcal{C} \simeq \mathrm{Comod}_{\mathrm{fd}}(\mathcal{O}(G)) = \mathrm{Rep}_k(G).$$

The construction of multiplication and symmetry makes this an equivalence of symmetric tensor categories, and its composite with the forgetful functor is ω. This proves neutral Tannaka reconstruction.

13.2 Remark. The pair $(\mathcal{C}, \omega)$ determines G. Keeping the fiber functor in the statement is essential: the category without a chosen fiber functor does not specify this identification. For comparison with the classical formulation, see [Mil12, Chapter X, Section 3] and [EGNO15, Section 5.4].

References

[BEO20] Dave Benson, Pavel Etingof, and Victor Ostrik. New incompressible symmetric tensor categories in positive characteristic, 2020.

[CEO23] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. On Frobenius exact symmetric tensor categories. *Annals of Mathematics*, 197(3):1235–1279, 2023.

[CEO24] Kevin Coulembier, Pavel Etingof, and Victor Ostrik. Incompressible tensor categories. *Advances in Mathematics*, 457:109935, 2024.

[CF24] Kevin Coulembier and Johannes Flake. Towards higher Frobenius functors for symmetric tensor categories, 2024. The course discussion follows the May 2024 version.

[Del02] Pierre Deligne. Catégories tensorielles. *Moscow Mathematical Journal*, 2(2):227–248, 2002.

[DM82] Pierre Deligne and James S. Milne. Tannakian categories. In *Hodge Cycles, Motives, and Shimura Varieties*, volume 900 of *Lecture Notes in Mathematics*, pages 101–228. Springer, 1982. Corrected version, 2022, with the original proposition numbering.

[EGNO15] Pavel Etingof, Shlomo Gelaki, Dmitri Nikshych, and Victor Ostrik. *Tensor categories*, volume 205 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2015.

[ENO05] Pavel Etingof, Dmitri Nikshych, and Viktor Ostrik. On fusion categories. *Annals of Mathematics*, 162(2):581–642, 2005.

[EO19] Pavel Etingof and Victor Ostrik. On semisimplification of tensor categories, 2019.

[Mil12] James S. Milne. Basic theory of affine group schemes, 2012. Available at www.jmilne.org/math/.

[Ost15] Victor Ostrik. On symmetric fusion categories in positive characteristic, 2015.

[Wat79] William. C. Waterhouse. *Introduction to Affine Group Schemes*. Springer Graduate texts in mathematics. Springer New York, NY, 1979.

[Wib25] Michael Wibmer. A remark on torsors under affine group schemes. *Transformation Groups*, 30:447–454, 2025.